
2018 Y6 H2 Chemistry Preliminary Exams Paper 2 (Suggested Solutions)

- 1(a) X is likely to be in Group 14 as there is a large jump from the 4th to 5th ionisation energies. This shows that the 5th electron is removed from an inner electron shell and requires more energy for removal as it is more strongly attracted to the nucleus. Hence, there are 4 valence electrons in X.

Since X has at least 10 electrons but fewer than 20 electrons, X is Si.

- 1(b) Mn and Fe are first-row transition metals. Although Fe has a higher nuclear charge (greater proton number) than Mn, Fe has more 3d electrons which provide more shielding between the nucleus and the outer 4s shell of electrons. This increase in shielding effect offsets the increase in nuclear charge and hence, the increase in effective nuclear charge from Mn to Fe is minimal, resulting in the first ionisation energies of Mn and Fe to be similar.

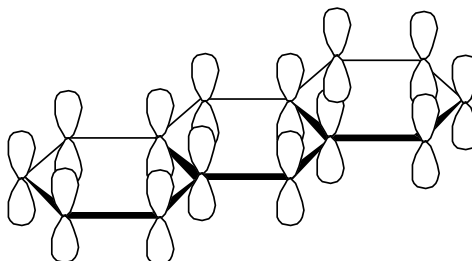
- 1(c) Ca^{2+} has a larger cationic radius than Cu^{2+} , thus Ca^{2+} has a lower charge density and weaker polarising power.

Consequently, there is lower extent of distortion of the electron cloud of the carbonate anion and hence covalent bonds within the carbonate anion are weakened to a lesser extent. Hence, CaCO_3 decomposes at a higher temperature.

- 2(a)(i) Anthracene is a cyclic molecule. All carbon atoms of anthracene are sp^2 hybridised and are trigonal planar.

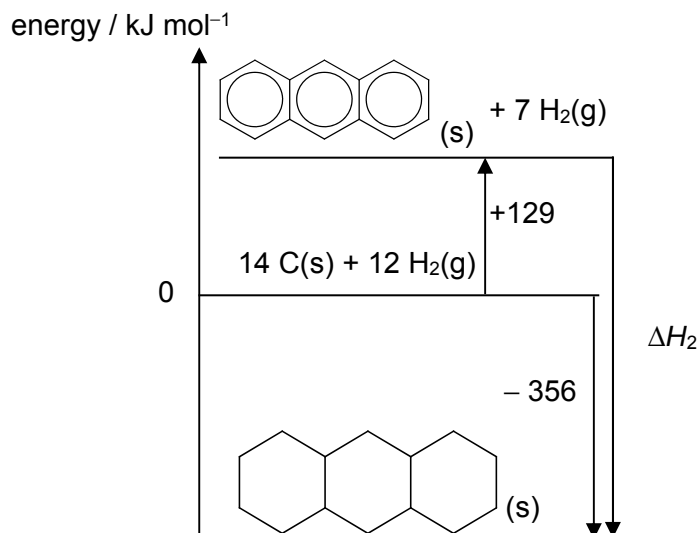
The unhybridised p-orbitals of the carbon atoms in anthracene can form a continuous p-orbital overlap, thus the π electrons are delocalised throughout the molecule.

Anthracene has $4(3) + 2 = 14$ π electrons where $n = 3$.



- 2(a)(ii) Enthalpy change of hydrogenation = $7(-118) = \underline{-826 \text{ kJ mol}^{-1}}$

2(a)(iii)



Enthalpy change of hydrogenation = $-356 - 129 = \underline{-485 \text{ kJ mol}^{-1}}$

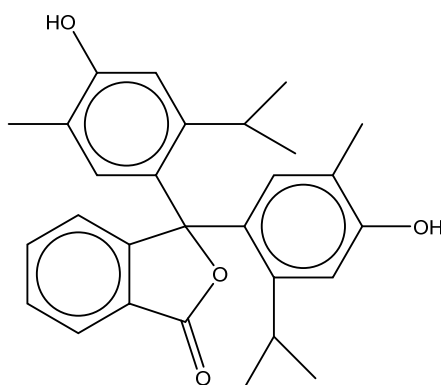
2(a)(iv) Resonance energy = $826 - 485 = \underline{341 \text{ kJ mol}^{-1}}$

2(b)(i) TEQ = $(0.029 \times 0.01) + (0.135 \times 0.01) + (0.204 \times 1.00) = 0.20564 = \underline{0.206}$

2(b)(ii) ECR = $\frac{0.20564}{70} \times 1.37 \times 10^{-3} = \underline{4.00 \times 10^{-6}}$

Since the ECR from the consumption of smoked fish exceeds the value set by USEPA, it will be a cancer risk for the 70 kg person.

2(c)(i)



2(c)(ii) Thymol reacts more readily as the alkyl groups of thymol are electron donating and increase the electron density on the benzene ring making it more susceptible to electrophilic substitution.

2(d) The carboxylate anion in **A** is more stable than the phenoxide anion in **B** as the negative charge on oxygen is dispersed over the two highly electronegative oxygen atoms resulting in two equivalent resonance structures in A as compared to B.

- 3(a)(i)** The stronger the base, the more available is its lone pair of electrons for donation to acid. Hence, a strong Lewis base will readily donate its lone pair to the electron deficient carboxyl C (Lewis acid) to reform the starting acid derivative. Thus, strong Lewis bases are poor leaving groups.

or

Strong Lewis bases donate electron pair more easily and are less likely to accept an electron pair. Thus, it is more difficult to break the C-Y bond to form Y⁻.

- 3(a)(ii)** Since pK_b of CH_3COO^- is more positive than NH_2^- , CH_3COO^- is a weaker base and is hence a better leaving group so that the reactivity of $(\text{CH}_3\text{CO})_2\text{O}$ is higher than that of CH_3CONH_2 . Reaction 1 will proceed with a faster rate than reaction 2.

or

The pK_b of CH_3O^- is less positive than that of CH_3COO^- but more positive than that of NH_2^- . Hence the product $\text{CH}_3\text{COOCH}_3$ is less reactive than $(\text{CH}_3\text{CO})_2\text{O}$ but more reactive than CH_3CONH_2 . Thus, reaction 1 is likely to have a higher yield, as the product is less reactive than the reactant so that the backward reaction is less likely to occur, resulting in higher yield.

- 3(b)(i)** Initial amount of $\text{CH}_3\text{OH} = \frac{20 \times 0.792}{32.0} = 0.495 \text{ mol}$

- 3(b)(ii)** $n_{\text{CH}_3\text{COOH}}$ in 5.00 cm^3 of eqm mixture = $n_{\text{NaOH}} = \frac{16.80}{1000} \times 0.500 = 0.00840 \text{ mol}$

$$n_{\text{CH}_3\text{COOH}} \text{ in } 40.00 \text{ cm}^3 \text{ of eqm mixture} = 0.0084 \times \frac{40.00}{5.00} = 0.0672 \text{ mol}$$

- 3(b)(iii)** $K_c = \frac{[\text{CH}_3\text{COOCH}_3][\text{H}_2\text{O}]}{[\text{CH}_3\text{COOH}][\text{CH}_3\text{OH}]}$

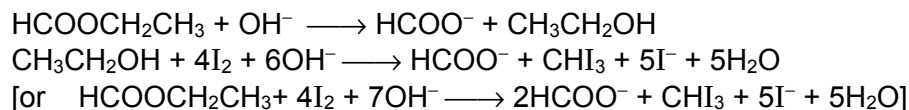
- 3(b)(iv)** $n_{\text{CH}_3\text{COOH}} \text{ reacted} = 0.350 - 0.0672 = 0.2828 \text{ mol}$

	$\text{CH}_3\text{OH(l)} + \text{CH}_3\text{COOH(l)} \rightleftharpoons \text{CH}_3\text{COOCH}_3\text{(l)} + \text{H}_2\text{O(l)}$			
Initial amt/ mol	0.495	0.350	0	0
Change in amt/ mol	-0.2828	-0.2828	+0.2828	+0.2828
Final amt/ mol	0.2122	0.0672	0.2828	0.2828

$$K_c = \frac{\left(\frac{0.2828}{V}\right)\left(\frac{0.2828}{V}\right)}{\left(\frac{0.2122}{V}\right)\left(\frac{0.0672}{V}\right)} = 5.61$$

- 3(b)(v)** The mixture was added to a large amount of ice-cold water to quench it. Cooling and diluting the reaction mixture slows down the reaction so that the equilibrium position would not shift during the titration and the equilibrium concentration of CH_3COOH can be determined.

- 3(c)** Warm the esters separately with aqueous alkaline iodine.
The ester that gives a yellow precipitate of CHI_3 is $\text{HCOOCH}_2\text{CH}_3$. No yellow precipitate for $\text{CH}_3\text{COOCH}_3$.

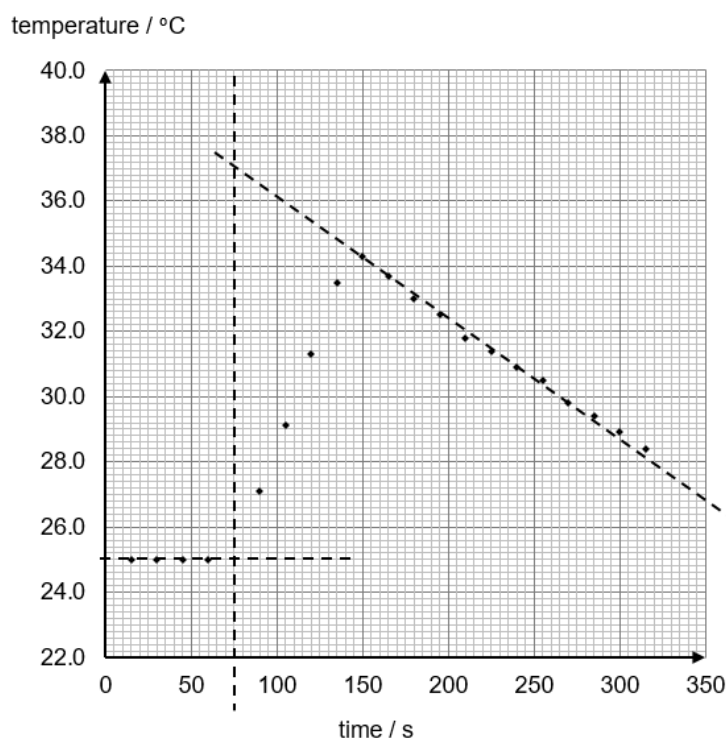


- 3(d)** Add dilute hydrochloric acid to the mixture. Stir and filter the mixture. **Q** is obtained as a salt in the filtrate while **P** is obtained as the residue.

Q is an amine and is basic. **Q** reacts with the acid to produce $(\text{C}_6\text{H}_5)_2\text{NH}_2^+$ (or salt) which forms strong ion-dipole interactions with water, resulting in its dissolution. Thus, **Q** dissolves in dilute hydrochloric acid and can be separated from **P** by filtration.

- 4(a)(i)** It is the energy change when one mole of the substance is completely dissolved in a solvent to form an infinitely dilute solution at 298 K and 1 bar.

- 4(a)(ii)** $\Delta T = 37.0 - 25.0 = \underline{+12.0\text{ }^\circ\text{C}}$



- 4(a)(iii)** Heat absorbed by water = $mc\Delta T = 100 \times 4.18 \times (12.0) = 5016\text{ J}$
Mr of anhydrous $\text{MgSO}_4 = 24.3 + 32.1 + 4(16.0) = 120.4$
Amt of anhydrous MgSO_4 used = $(7.500 / 120.4) = 0.06229\text{ mol}$
 $\Delta H_{\text{sol}}^\ominus = - (5016 / 0.06229) = -80527\text{ J mol}^{-1} = \underline{-80.5\text{ kJ mol}^{-1}}$

4(a)(iv) The experimental value is less exothermic than the actual value.

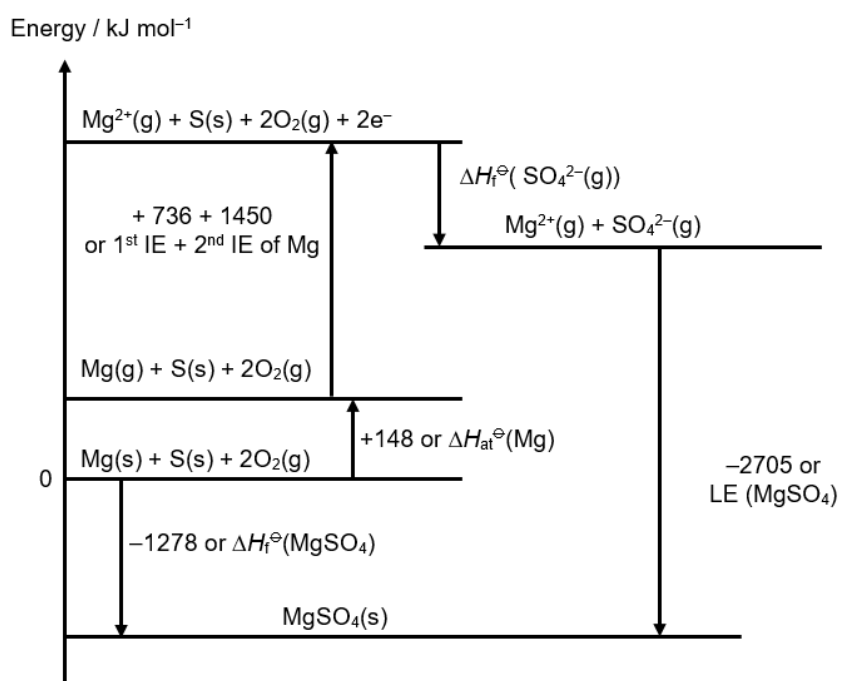
The heat capacity of cup/calorimeter was not taken into consideration. Account for the heat capacity of the cup/calorimeter by calibrating the calorimeter using a sample with known ΔH_{soln} .

or

The solution formed was not sufficiently dilute. Use a larger volume of water or use less solid.

4(a)(v) $\Delta H_r^\ominus = -88.0 - (+16.0) = \underline{-104 \text{ kJ mol}^{-1}}$

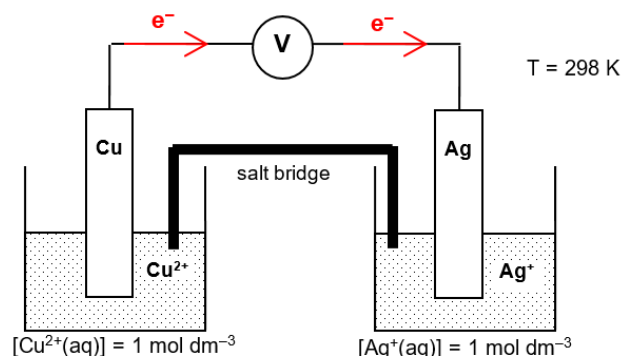
4(b)



$$-1278 = +148 + 736 + 1450 + \Delta H_f^\ominus(\text{SO}_4^{2-}(\text{g})) - 2705$$

$$\Delta H_f^\ominus(\text{SO}_4^{2-}(\text{g})) = \underline{-907 \text{ kJ mol}^{-1}}$$

5(a)(i)



5(a)(ii) $E^\ominus_{\text{cell}} = E^\ominus_{\text{cathode}} - E^\ominus_{\text{anode}}$
 $= +0.80 - (+0.34)$
 $= \underline{+0.46 \text{ V}}$

- 5(a)(iii) Upon addition of NH_3 , blue ppt forms.
 $\text{Cu}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightleftharpoons \text{Cu}(\text{OH})_2(\text{s})$ blue ppt

Upon addition of excess NH_3 , ppt dissolves and deep blue solution is formed.
 $\text{Cu}^{2+}(\text{aq}) + 4\text{NH}_3(\text{aq}) \rightleftharpoons [\text{Cu}(\text{NH}_3)_4]^{2+}(\text{aq})$ deep blue complex

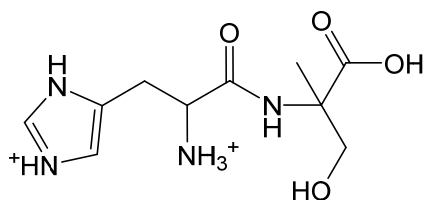
- 5(a)(iv) Upon addition of NH_3 , $[\text{Cu}^{2+}]$ decreases such that it becomes less than 1 mol dm^{-3} (i.e. conditions are no longer standard). Position of equilibrium for $\text{Cu}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Cu}$ shifts to the left.

E_{anode} in (a)(ii) decreases and hence E_{cell} becomes more positive.

- 5(b) $\Delta G^{\ominus} = -nFE^{\ominus}_{\text{cell}} = -(2)(96500)(+1.60) = \underline{-309 \text{ kJ mol}^{-1}}$

- 6(a) ams and msr as they do not follow the general formula $\text{NH}_2\text{CH(R)COOH}$.

- 6(b)



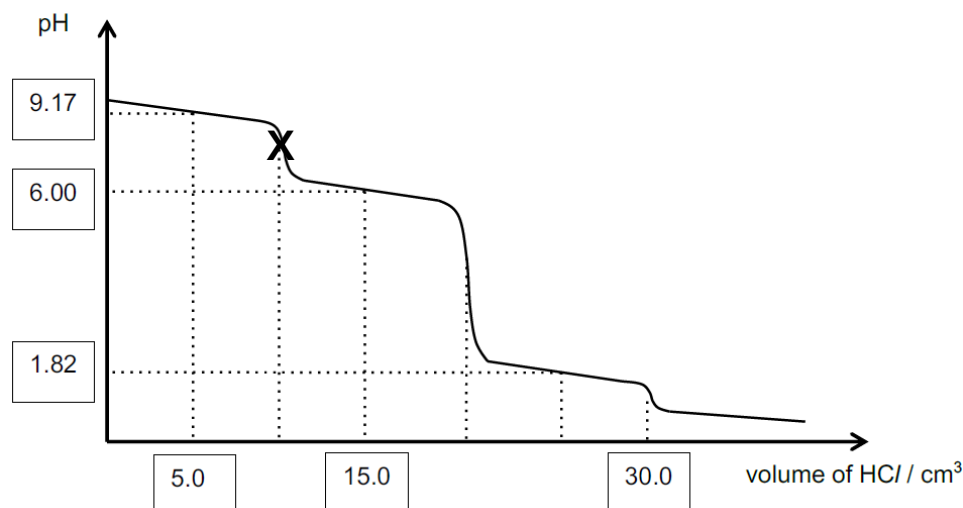
- 6(c)(i) $\text{p}K_{\text{b}}$ of $\text{N}^{\textcircled{1}}$ = $14 - 9.17 = \underline{4.83}$
 $\text{p}K_{\text{b}}$ of $\text{N}^{\textcircled{3}}$ = $14 - 6.00 = \underline{8.00}$

- 6(c)(ii) sp^3

- 6(c)(iii) The lone pair on $\text{N}^{\textcircled{1}}$ is located in the sp^3 hybrid orbital which has less s character than the lone pair on $\text{N}^{\textcircled{3}}$ which is located in the sp^2 hybrid orbital. Therefore, $\text{N}^{\textcircled{1}}$ lone pair is less strongly attracted to the nucleus and is more available for coordination with a proton.

- 6(c)(iv) The unhybridised p orbital on $\text{N}^{\textcircled{2}}$ can overlap with the unhybridised p orbitals of the other atoms in the ring. Therefore the lone pair is delocalised and unavailable for coordination with a proton.

- 6(d) &
6(f)(ii)

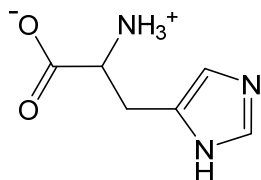


6(e) $n(\text{HCl}) \text{ added} = n(\text{conjugate acid}) \text{ formed} = 6/1000 \times 0.1 = 0.0006 \text{ mol}$
 $n(\text{fully deprotonated his}) \text{ remaining} = (10/1000 \times 0.1) - 0.0006 = 0.0004 \text{ mol}$

$$\begin{aligned} \text{pH} &= \text{p}K_{\text{a}} + \lg ([\text{fully deprotonated his}] / [\text{conjugate acid}]) \\ &= 9.17 + \lg (0.0004 / 0.0006) \\ &= \underline{8.99} \end{aligned}$$

6(f)(i)

point Y



point Z

